SG No. 196 T^2 $F23$ [cubic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$, $+\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$

Table 1: Wyckoff site: `4a`, site symmetry: `23.`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `4b`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 4: Wyckoff site: `4d`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 5: Wyckoff site: `16e`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,9]`
2	$[-x,\ -x,\ x]$	`[2,7,12]`
3	$[-x,\ x,\ -x]$	`[3,8,10]`
4	$[x,\ -x,\ -x]$	`[4,6,11]`

Table 6: Wyckoff site: `24f`, site symmetry: `2..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4]`
2	$[-x,\ 0,\ 0]$	`[2,3]`
3	$[0,\ x,\ 0]$	`[5,8]`
4	$[0,\ -x,\ 0]$	`[6,7]`
5	$[0,\ 0,\ x]$	`[9,12]`
6	$[0,\ 0,\ -x]$	`[10,11]`

Table 7: Wyckoff site: `24g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,4]`
2	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,3]`
3	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[5,8]`
4	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	`[6,7]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	`[9,12]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	`[10,11]`

Table 8: Wyckoff site: `48h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$[x,\ -y,\ -z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[z,\ -x,\ -y]$	`[6]`
7	$[-z,\ -x,\ y]$	`[7]`
8	$[-z,\ x,\ -y]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$[-y,\ z,\ -x]$	`[10]`
11	$[y,\ -z,\ -x]$	`[11]`
12	$[-y,\ -z,\ x]$	`[12]`